

# Alexander Tereshin

## AI Research Engineer

Agentic Systems · RAG · Production LLM

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AI Research Engineer building **agentic systems** that hold up in production. Architected a corporate **multi-agent AI platform** at a major bank, where autonomous agents generated product hypotheses with a **67% acceptance rate** from product teams; now developing **agentic VLM-based RAG** that outperforms published SOTA baselines on financial-document QA. Background in **classical ML** and **NLP**: forecasting, classification pipelines, and large-scale A/B experimentation in e-commerce.

### EXPERIENCE

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#### NLP Researcher AIRI

May 2026 – Present

- Designing an **agentic VLM-based RAG** system for text-and-table document QA over real-world **financial reports**, outperforming published SOTA baselines in numerical-reasoning accuracy (**Number Match**) on **T<sup>2</sup>-RAGBench**; first-author conference paper in preparation.

#### Teaching and Research Assistant Higher School of Economics (HSE University)

Sep 2025 – Present

- Teaching assistant for graduate **Basic & Advanced NLP** courses (300+ students); designed curriculum and examined 30+ students on topics from **Transformer internals** to **Multi-Agent Systems**;
- Supervised low-level LLM implementation - manual **Qwen2.5** architecture (**RoPE**, **RMSNorm**, **SwiGLU**) for speculative decoding, plus **PEFT/QLoRA** fine-tuning and **GPTQ** quantization.

#### Machine Learning Engineer Raiffeisenbank

Jun 2024 – May 2026

- Architected and delivered a corporate **multi-agent AI platform** where autonomous agents explore user data to **generate product hypotheses**, achieving a **67% acceptance rate** from product teams for further testing. Equipped agents with an **Agentic RAG** tool (**FastMCP**) for banking-domain context and a **Zero-Touch QA agent (Ragas)** for automated evaluation, served by an internal **vLLM** stack (**GLM** for agent reasoning, **Qwen-2.5-Coder** for code generation) in isolated Micro Sandboxes;
- Scaled data labeling throughput 10x** (to **2,000 labels/day**) at **92% accuracy** by engineering an async system (**Litestar**, **FastStream**, **RabbitMQ**) driven by **Qwen2.5 Few-Shot CoT**;
- Designed and deployed 4 production batch pipelines (**PySpark**, **HDFS**, **Airflow**) for a **LightGBM** model (**81% F1**), implementing full-stack observability (**Evidently AI**, **Prometheus**, **Grafana**) to monitor data drift and model degradation;
- Cut HR policy query resolution time by 99%** (hours to seconds) with a Hybrid RAG system (**Milvus**, **BM25** + **BGE-M3**) automating **80%** of basic responses at **80 ms p95 retrieval latency**.

#### Data Scientist — Retail & E-commerce

Jan 2023 – Jun 2024

Sportmaster (Jan–Jun 2023) · VseInstrumenti.ru (Jun 2023 – Feb 2024) · Leroy Merlin (Feb–Jun 2024)

- VseInstrumenti.ru**: Led Data Science for three core products on a **12M+ MAU** platform; ran **40+ online A/B experiments** with advanced variance reduction (**Stratification**, **Delta Method**), directly driving a **\$25K** monthly revenue increase;
- Leroy Merlin**: Built churn forecasting with **SARIMA** and interpreted a **CatBoost** model via **SHAP** to surface churn drivers and retention insights for sales and product teams;
- Sportmaster**: Built a production ML classification pipeline for automated warehouse safety-risk detection.

#### Career Transition into Machine Learning

Feb 2022 – Jan 2023

Full-time self-study

- Full-time transition into ML: statistics, Python, and core algorithms, leading to admission to the HSE Master's program in Machine Learning and High Load Systems.

## EDUCATION

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### Higher School of Economics (HSE University)

Master's degree in Machine Learning and High Load Systems

Sep 2023 – Jun 2025

*GPA: 8.29/10*

### Saint Petersburg State University of Civil Aviation

Bachelor's degree in Maintenance of Aircraft and Engines

Sep 2016 – Jun 2021

## EARLIER EXPERIENCE

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### Aviation Engineering & Operations

2018 – 2022

- Progressed from Airframe & Powerplant Engineer to **Lead Engineer** in business aviation and then **Production Shift Manager**, leading cross-functional teams of **60+ people** across 30+ safety-critical Boeing / Airbus maintenance projects; managed data-driven resource allocation and AOG (aircraft-on-ground) recovery under strict reliability constraints.

## SKILLS

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**Programming & ML:** Python, PyTorch, Transformers, scikit-learn, LightGBM, CatBoost

**Agentic & LLM:** Agentic Workflows, LangChain, LangGraph, Pydantic AI, MCP, RAG, PEFT/QLoRA, GPTQ

**Evaluation & Statistics:** Experimental Design, A/B Testing, Variance Reduction, Drift Monitoring, Ragas

**MLOps & Orchestration:** Airflow, Kubernetes, Docker, CI/CD, MLflow, vLLM, Evidently AI, Prometheus, Grafana

**Data & Databases:** PySpark, SQL, Hadoop, Vector DBs (Milvus), MongoDB, AWS S3

**Backend & Tools:** FastAPI, Litestar, Kafka, RabbitMQ, Redis, Bash, Linux, Git, Pytest

**Languages:** English (B2 — Professional Working Proficiency), Russian (Native)